

Quantum Machine Learning for Calorimeter Simulation

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PRC Meetings

Generative Modeling for Calorimeter Data

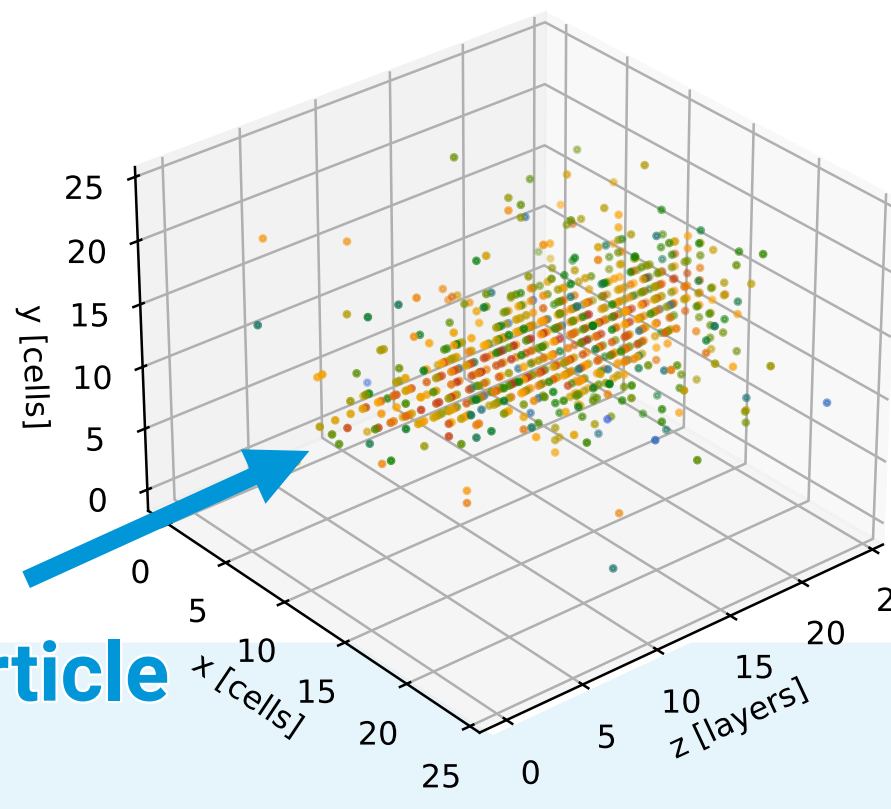
Use case: Simulation of shower images from calorimeters with QML.

Dataset: Shower images recorded by particles within the energy range of [225, 275] GeV.

Potential advantages with QML:

- Better generalization capabilities
- Fewer parameters needed
- Higher expressivity

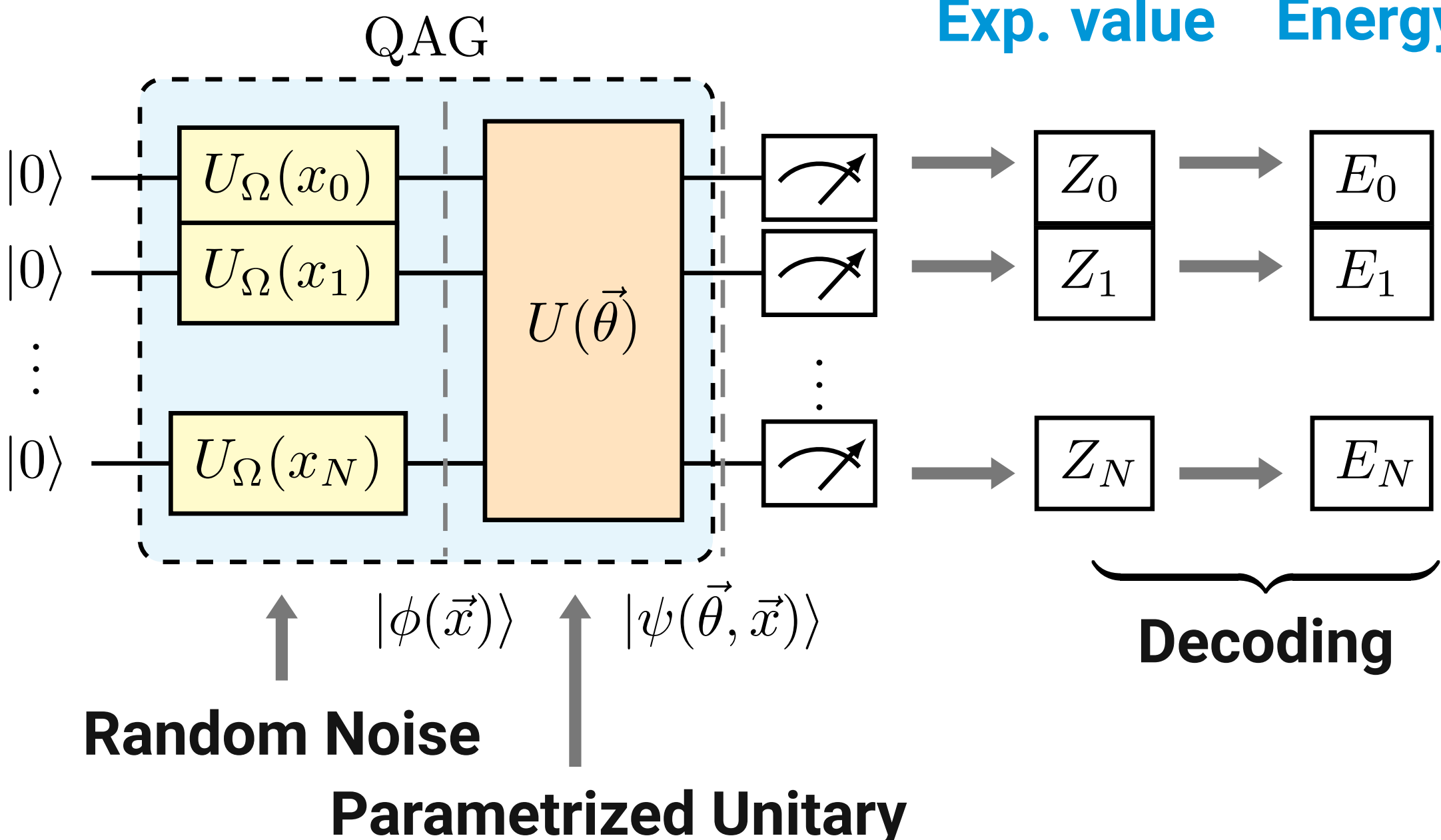
Sought advantages in learning complex distributions more efficiently



Particle

Quantum Angle Generator

- Encoding:** Classical pixel energy into a quantum state, in form of a rotational angle of a qubit.
- Decoding:** Translate quantum state into an energy by measuring qubit multiple times (n_{shots}).



QAG

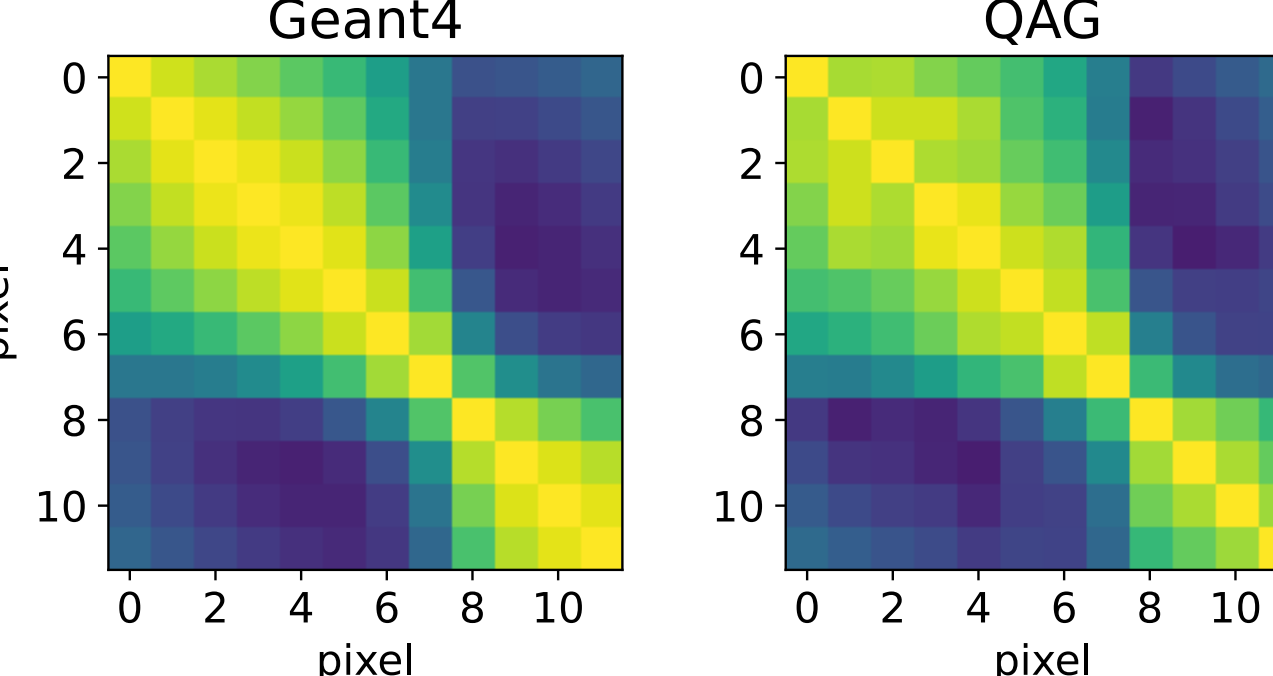
Exp. value Energy

Decoding

Random Noise

Parametrized Unitary

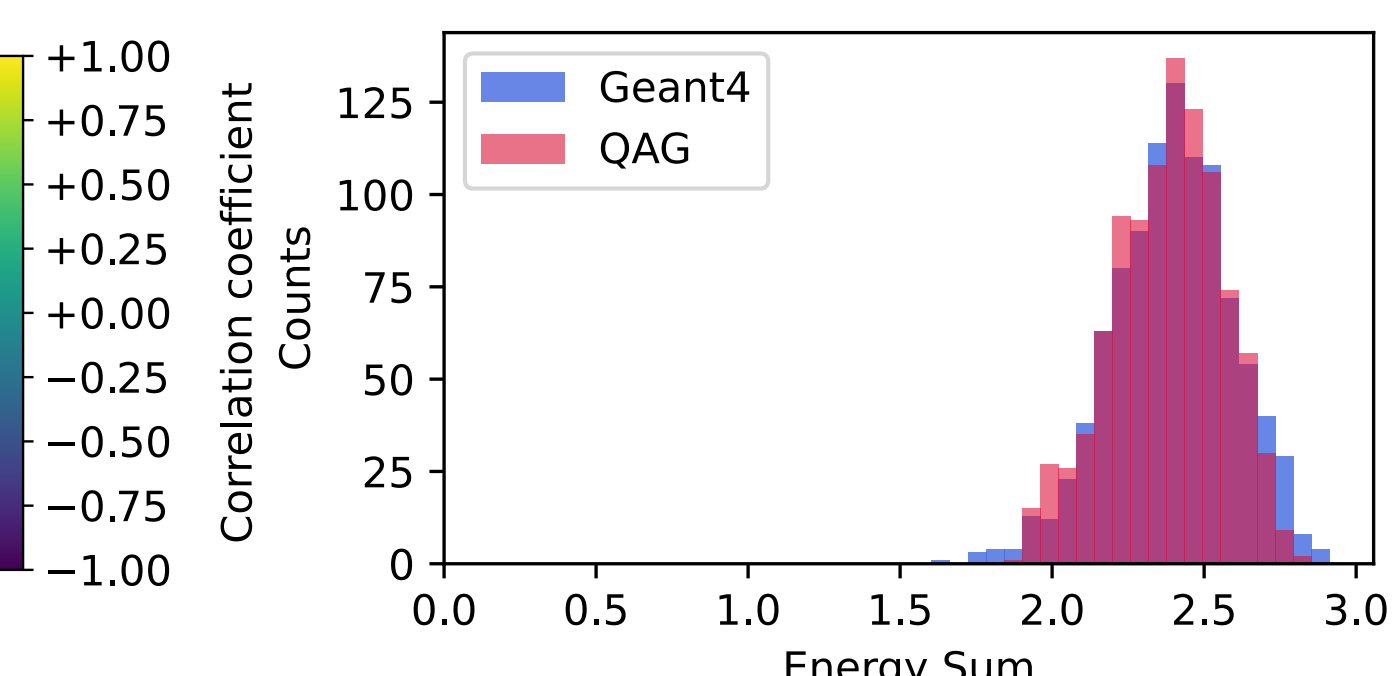
Correlations:



Geant4

QAG

Total energy:

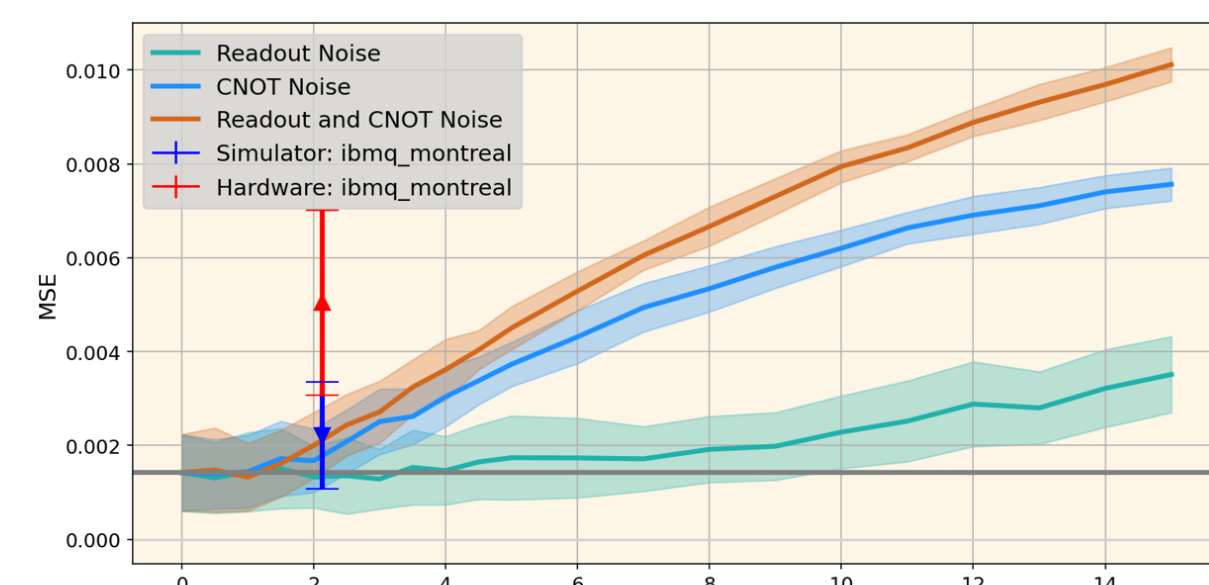


Geant4

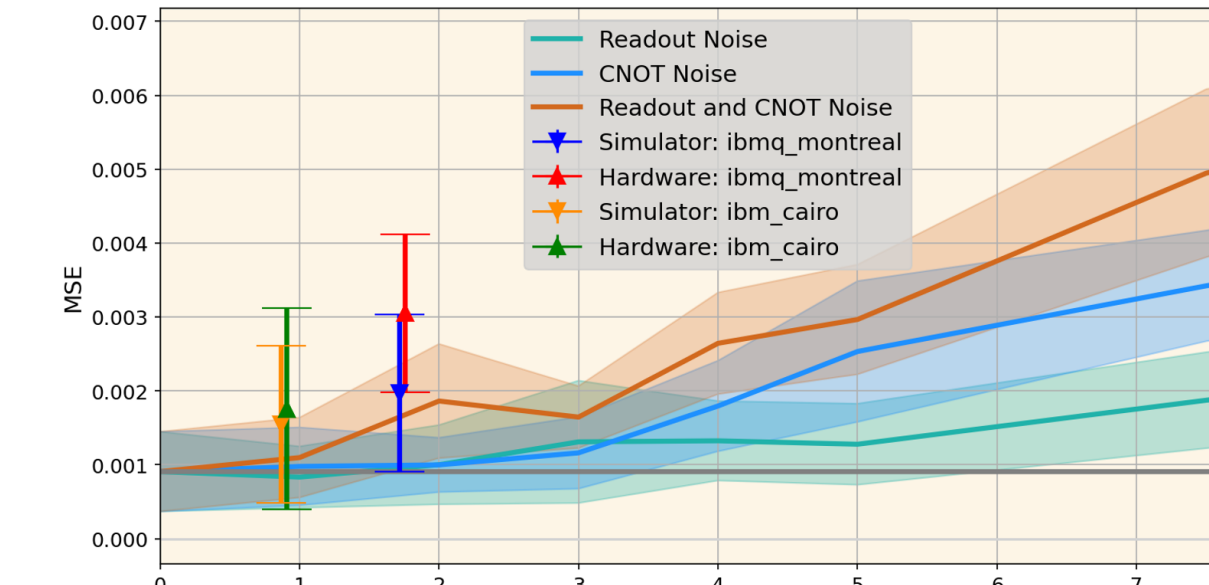
QAG

Noise study:

Inference: Noise only during inference



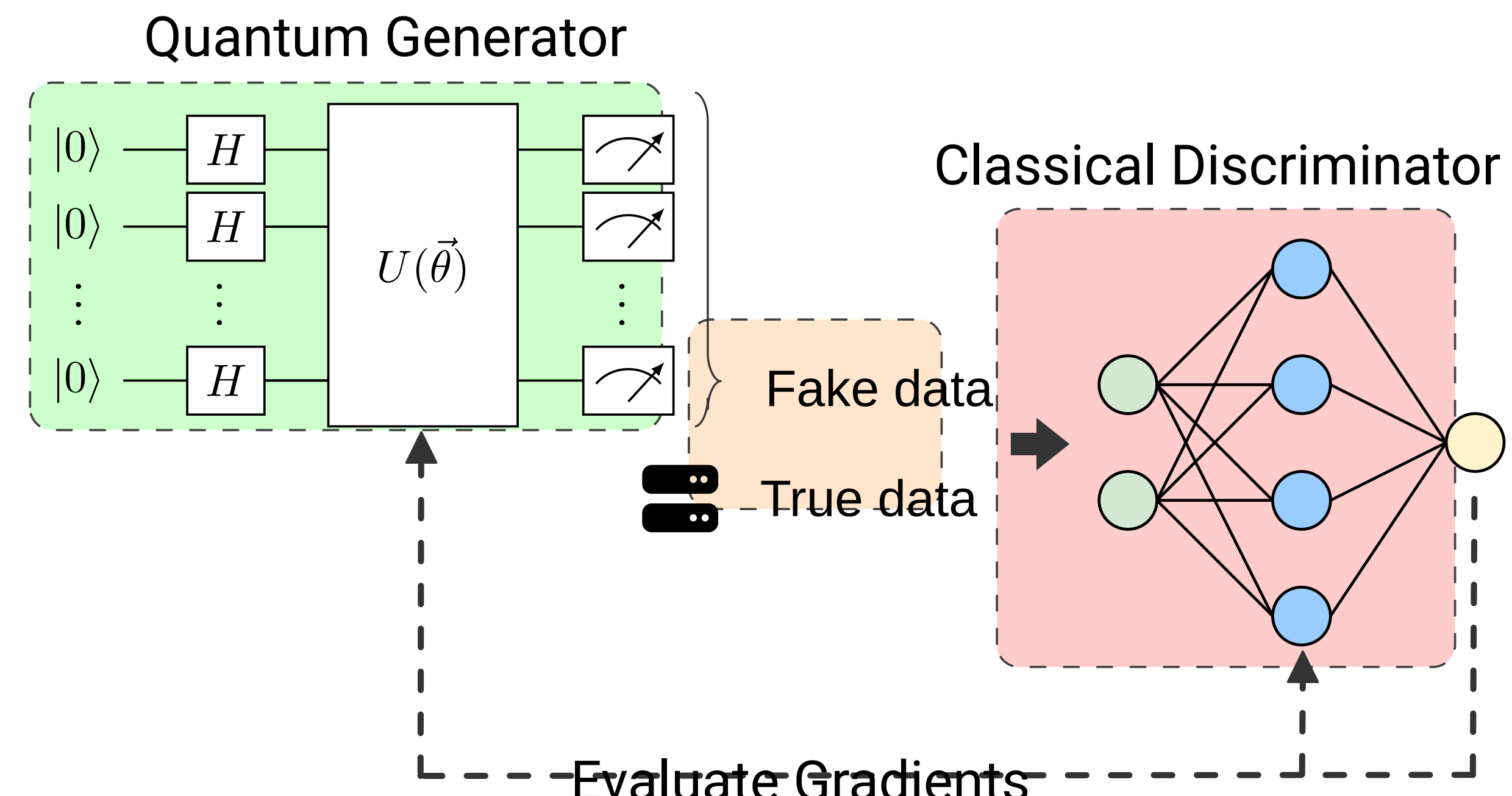
Training: Noise during training phase



- Smooth training with fast convergence
- 20x less parameters compared to classical counterparts
- Adaptation to the underlying noise

Quantum Generative Adversarial Network

- Hybrid Design:** Quantum Generator + Classical Discriminator



Quantum Generator

Classical Discriminator

Fake data

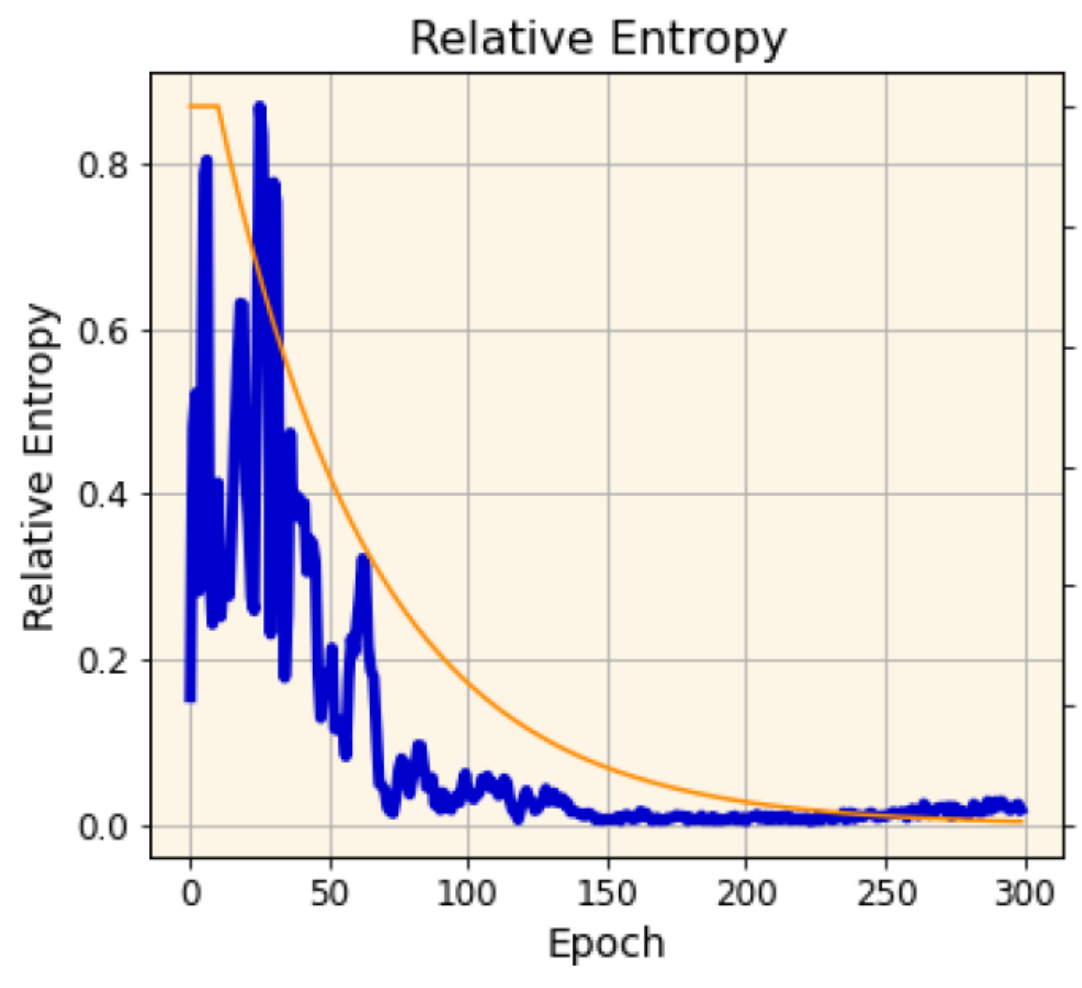
True data

Evaluate Gradients & Update parameters

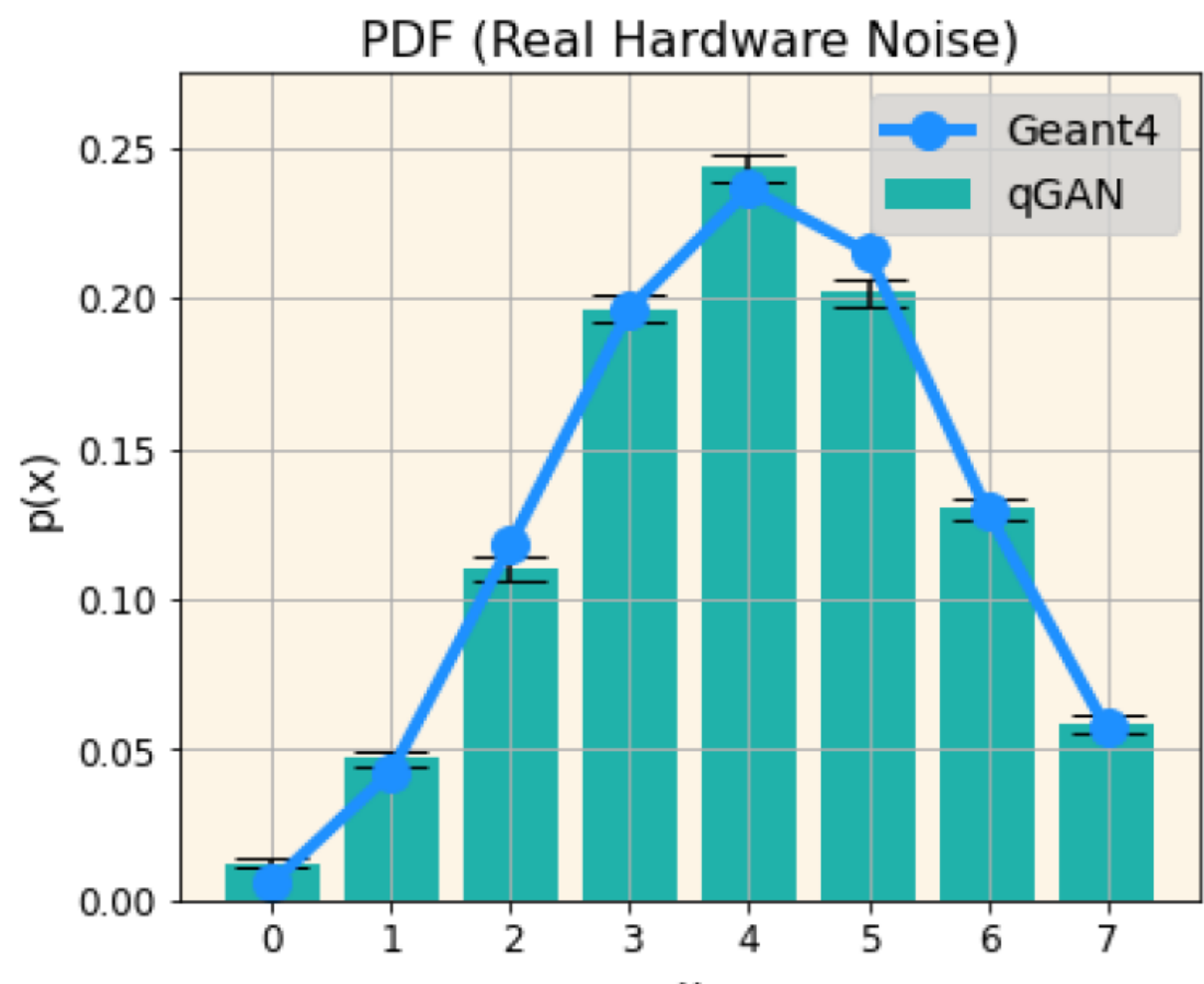
Performances

Generator run on *ibmq athens* quantum device.

Training:



Energy PDF:



★ Higher scalability through amplitude encoding

Implementation on real hardware

Testing the architectures on real quantum processors under different noise types and levels.



IBM

IQM

eleQtron

Superconducting

Superconducting

Ion-Trap

Next steps

- ✓ Checklist for developing new models:
 - Does the model suffer from Barren Plateaus?
 - Quantum generative models suffer from BP [3]. Training might be difficult scaling up.
 - Is the model Classically simulable? [4]
 - The same algorithm can be easily run on a classical computer
 - Is the model implementable on a real device?
 - Short coherence time + noise resistance
- + Bonus Point: if training can be done classically while keeping the model non-classically simulable [5]



- [3] Larocca et al., A review of barren plateaus in variational quantum computing, (2024)
- [4] Angrisani et al., Classically estimating observables of noiseless quantum circuits, (2024)
- [5] Recio-Armengol et al., Train on classical, deploy on quantum: scaling generative quantum machine learning to a thousand qubits, (2025)

References

[1] Florian Rehm: *Deep learning and quantum generative models for high energy physics calorimeter simulations*, RWTH Aachen University, PhD Dissertation (2023)

[2] Florian Rehm, Sofia Vallecorsa, Kerstin Borrás, Dirk Krücker, Michele Grossi, Valle Varo: *Precise Image Generation on Current Noisy Quantum Computing Devices*, Quantum Science and Technology (2023)